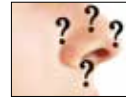




### Temperatures soaring

The Potomac River in Washington has hit a record high temperature of 94-degrees F or 34-degrees C. <https://digit.in/aug19-05>



### Concussions cause loss of smell

A new study has found that even mild concussions can cause difficulty in identifying smells the next day. <https://digit.in/aug19-06>

and KEK. Theoretically, there could even be pentaquarks which are exotic hadrons made up of five quarks. Such particles may even occur in nature because of supernovae. What scientists know confidently enough is that there are existing exotic hadrons which are particles that do not fit into the traditional quark model. Essentially what this means is that unusual decay events have been recorded, but the nature of the particle and its composition is unclear. One of the problems is that the resulting observations can be explained with the particles that we already know of. It is now a matter of measuring them and finding out their exact properties. Researchers expect to discover exotic hadrons at LHC within the next few years.

### PREON

The atom was considered to be the fundamental building block of matter, till we split it open and found protons, neutrons and electrons within them. Neutrons and protons, along with other subatomic particles are made up of quarks, that come in six flavors. These are currently considered to be elementary particles, particles that cannot be broken down further. Quarks are never observed in isolation, and always exist as a component of a hadron. Leptons such as the electron are also elementary particles. We know of over 50 subatomic particles now, and the preon is an effort to reduce the number of fundamental particles. The idea is to reduce the number of building blocks in nature to about three, similar to how atoms could be understood with just a proton, a neutron and an electron. One of the signs pointing to a more fundamental physics than the Standard Model, is that quarks have been observed to decay into other quarks, which indicates that they cannot be the fundamental building blocks of nature. There have been many preon models over the years, but a lot of them pre-dated the discovery of the Higgs Boson, and attempted to explain away the lack of such a particle. Current efforts are centered around explaining

some of the peculiarities or aberrations in the Standard Model.

### QUANTUM BLACK HOLES

We do not blame you if you directly jumped to this section looking at the sub head. Yes, the LHC can produce miniature black holes, and no, it will not eat up the whole Earth. The black holes created are expected to rapidly evaporate because of Hawking Radiation, or pass harmlessly through much denser matter around it, like a rock sinking through water. There are simulations of what a decaying black hole would look like to the LHC experiments, and these are under-



Beyond the Standard Model

standably spectacular events with jets of many particles, including electrons and muons. Such an event would produce an unusually large amount of particles, not seen in most collisions and subsequent decays. Theoretically, a collision between two protons can produce such a quantum black hole. However, the implication is even more exciting - that the universe has additional, hidden dimensions. The implications of such a discovery are immense, and would instantly open up a range of possibilities how men can use these extra dimensions. These uses include faster ways of navigating the spacetime, and a new source of cheap energy. These microscopic black holes are predicted to exist if gravity is

stronger, or leaking out into additional spatial dimensions that we cannot see, which brings us to the next theoretical particle that scientists are hunting for.

### GRAVITON

Within the standard model, there are force-carrying particles associated with three of the four fundamental forces. Electromagnetism has photons, strong force has gluons and the weak force is carried by the W and Z bosons. The standard model leaves out gravity because of how weak it is compared to all the other forces, and the effects of gravity on such tiny scales are negligible, or not easily measured by instruments. A button magnet will jump up to another, against the gravitational pull of the entire Earth. The theory of relativity explains gravity well enough. Researchers looking into so called quantum gravity, are hunting for the force carrying particle associated with gravity known as the graviton. The gravitational waves detected by the LIGO and Virgo observatories from the collisions of black holes and neutron stars, may actually be a massless form of the elusive graviton. One of the explanations for why gravity is so weak, is that its effects could be leaking away into spatial dimensions other than the three that is known to science. Finding unusually massive particles can also hint at the existence of these extra dimensions. Theoretically, physicists anticipate that a graviton would decay into a pair of Z bosons, which further decay into other particles. One of the Z bosons decay to a pair of muons, and the other Z boson would decay to a jet of particles. The thing is even if a graviton was created within a particle accelerator, the resulting cluster of particles that will eventually be detected can be explained by particles other than the graviton. There are various other particles that can decay into the same particles, so hunting the graviton would involve observing multiple such events and estimating the mass of the original particle. The outlook for this particle also looks good.